QMM_Module 8 Assignment_Andrew Claxton

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```
library(Benchmarking)
```

```
## Loading required package: lpSolveAPI
```

```
## Loading required package: ucminf
```

```
## Loading required package: quadprog
```

```
library(knitr)

# Facility names
facilities <- paste("Facility", 1:6)

# Inputs: staff hours and supplies
X <- cbind(
  Staff_Hours = c(150, 400, 320, 520, 350, 320),
  Supplies    = c(0.2, 0.7, 1.2, 2.0, 1.2, 0.7)
)
```

```
# Outputs, in thousands
Y <- cbind(
  Third_Party_Patient_Days = c(14, 14, 42, 28, 19, 14),
  Private_Patient_Days     = c(3.5, 21, 10.5, 42, 25, 15)
)
rownames(X) <- facilities
rownames(Y) <- facilities

#Assignment model names
model_codes <- c(
  FDH = "fdh",
  VRS = "vrs",
  DRS = "drs",
  IRS = "irs",
  CRS = "crs",
  FRH = "add"
)
```

```
#Objects to store results

models <- list()

# Facility by Model efficiency table.
```

```r
efficiency_table <- data.frame(
  Facility = facilities
)

# Peer and lambda table
all_peer_lambdas <- data.frame(
  Model = character(),
  Facility_Evaluated = character(),
  Peer_Facility = character(),
  Lambda = numeric()
)
```

```r
#Running every model and printing result

for (model_name in names(model_codes)) {
  cat("\n")
  cat("=====================================================\n")
  cat("MODEL:", model_name, "\n")
  cat("RTS CODE:", model_codes[[model_name]], "\n")
  cat("=====================================================\n")

  model_fit <- dea(
    X,
    Y,
    RTS = model_codes[[model_name]]
  )

  # Save all results
  models[[model_name]] <- model_fit

#Print efficiency scores and add scores to final efficiency table
  model_efficiency <- data.frame(
    Facility = facilities,
    Efficiency = round(model_fit$eff, 4)
  )

  cat("\nEfficiency Scores:\n")
  print(model_efficiency)

  # Add this model's scores to the final efficiency table.
  efficiency_table[[model_name]] <- round(model_fit$eff, 4)

#Print peers and lambdas

lambda_matrix <- model_fit$lambda

  # Force readable row and column names.
  rownames(lambda_matrix) <- facilities
  colnames(lambda_matrix) <- facilities

  # Convert lambda matrix into a long table.
  peer_lambda_table <- as.data.frame(as.table(lambda_matrix))
```

```
  names(peer_lambda_table) <- c(
    "Facility_Evaluated",
    "Peer_Facility",
    "Lambda"
  )

  # Keep only actual peers: lambda values greater than zero.
  peer_lambda_table <- subset(
    peer_lambda_table,
    Lambda > 0.000001
  )

  # Add model name and clean the output.
  peer_lambda_table$Model <- model_name
  peer_lambda_table$Lambda <- round(peer_lambda_table$Lambda, 4)

  peer_lambda_table <- peer_lambda_table[
    , c("Model", "Facility_Evaluated", "Peer_Facility", "Lambda")
  ]

  cat("\nPeers and Nonzero Lambdas:\n")
  print(peer_lambda_table)

  # Add this model's peer/lambda results to the combined table.
  all_peer_lambdas <- rbind(
    all_peer_lambdas,
    peer_lambda_table
  )
}
```

```
##
## ====================================================
## MODEL: FDH
## RTS CODE: fdh
## ====================================================
##
## Efficiency Scores:
##     Facility Efficiency
## 1 Facility 1          1
## 2 Facility 2          1
## 3 Facility 3          1
## 4 Facility 4          1
## 5 Facility 5          1
## 6 Facility 6          1
##
## Peers and Nonzero Lambdas:
##    Model Facility_Evaluated Peer_Facility Lambda
## 1    FDH         Facility 1    Facility 1      1
## 8    FDH         Facility 2    Facility 2      1
## 15   FDH         Facility 3    Facility 3      1
## 22   FDH         Facility 4    Facility 4      1
## 29   FDH         Facility 5    Facility 5      1
## 36   FDH         Facility 6    Facility 6      1
```

```
##
## ====================================================
## MODEL: VRS
## RTS CODE: vrs
## ====================================================
##
## Efficiency Scores:
##              Facility Efficiency
## Facility 1 Facility 1     1.0000
## Facility 2 Facility 2     1.0000
## Facility 3 Facility 3     1.0000
## Facility 4 Facility 4     1.0000
## Facility 5 Facility 5     1.0000
## Facility 6 Facility 6     0.8963
##
## Peers and Nonzero Lambdas:
##    Model Facility_Evaluated Peer_Facility Lambda
## 1    VRS         Facility 1    Facility 1 1.0000
## 6    VRS         Facility 6    Facility 1 0.4014
## 8    VRS         Facility 2    Facility 2 1.0000
## 12   VRS         Facility 6    Facility 2 0.3423
## 15   VRS         Facility 3    Facility 3 1.0000
## 22   VRS         Facility 4    Facility 4 1.0000
## 29   VRS         Facility 5    Facility 5 1.0000
## 30   VRS         Facility 6    Facility 5 0.2563
##
## ====================================================
## MODEL: DRS
## RTS CODE: drs
## ====================================================
##
## Efficiency Scores:
##              Facility Efficiency
## Facility 1 Facility 1     1.0000
## Facility 2 Facility 2     1.0000
## Facility 3 Facility 3     1.0000
## Facility 4 Facility 4     1.0000
## Facility 5 Facility 5     0.9775
## Facility 6 Facility 6     0.8675
##
## Peers and Nonzero Lambdas:
##    Model Facility_Evaluated Peer_Facility Lambda
## 1    DRS         Facility 1    Facility 1 1.0000
## 5    DRS         Facility 5    Facility 1 0.2000
## 6    DRS         Facility 6    Facility 1 0.3429
## 8    DRS         Facility 2    Facility 2 1.0000
## 11   DRS         Facility 5    Facility 2 0.0805
## 12   DRS         Facility 6    Facility 2 0.3950
## 15   DRS         Facility 3    Facility 3 1.0000
## 22   DRS         Facility 4    Facility 4 1.0000
## 23   DRS         Facility 5    Facility 4 0.5383
## 24   DRS         Facility 6    Facility 4 0.1311
##
## ====================================================
```

```
## MODEL: IRS
## RTS CODE: irs
## ====================================================
##
## Efficiency Scores:
##              Facility Efficiency
## Facility 1 Facility 1     1.0000
## Facility 2 Facility 2     1.0000
## Facility 3 Facility 3     1.0000
## Facility 4 Facility 4     1.0000
## Facility 5 Facility 5     1.0000
## Facility 6 Facility 6     0.8963
##
## Peers and Nonzero Lambdas:
##    Model Facility_Evaluated Peer_Facility Lambda
## 1    IRS         Facility 1    Facility 1 1.0000
## 6    IRS         Facility 6    Facility 1 0.4014
## 8    IRS         Facility 2    Facility 2 1.0000
## 12   IRS         Facility 6    Facility 2 0.3423
## 15   IRS         Facility 3    Facility 3 1.0000
## 22   IRS         Facility 4    Facility 4 1.0000
## 29   IRS         Facility 5    Facility 5 1.0000
## 30   IRS         Facility 6    Facility 5 0.2563
##
## ====================================================
## MODEL: CRS
## RTS CODE: crs
## ====================================================
##
## Efficiency Scores:
##              Facility Efficiency
## Facility 1 Facility 1     1.0000
## Facility 2 Facility 2     1.0000
## Facility 3 Facility 3     1.0000
## Facility 4 Facility 4     1.0000
## Facility 5 Facility 5     0.9775
## Facility 6 Facility 6     0.8675
##
## Peers and Nonzero Lambdas:
##    Model Facility_Evaluated Peer_Facility Lambda
## 1    CRS         Facility 1    Facility 1 1.0000
## 5    CRS         Facility 5    Facility 1 0.2000
## 6    CRS         Facility 6    Facility 1 0.3429
## 8    CRS         Facility 2    Facility 2 1.0000
## 11   CRS         Facility 5    Facility 2 0.0805
## 12   CRS         Facility 6    Facility 2 0.3950
## 15   CRS         Facility 3    Facility 3 1.0000
## 22   CRS         Facility 4    Facility 4 1.0000
## 23   CRS         Facility 5    Facility 4 0.5383
## 24   CRS         Facility 6    Facility 4 0.1311
##
## ====================================================
## MODEL: FRH
## RTS CODE: add
```

```
## ======================================================
##
## Efficiency Scores:
##              Facility Efficiency
## Facility 1 Facility 1          1
## Facility 2 Facility 2          1
## Facility 3 Facility 3          1
## Facility 4 Facility 4          1
## Facility 5 Facility 5          1
## Facility 6 Facility 6          1
##
## Peers and Nonzero Lambdas:
##    Model Facility_Evaluated Peer_Facility Lambda
## 1    FRH         Facility 1    Facility 1      1
## 8    FRH         Facility 2    Facility 2      1
## 15   FRH         Facility 3    Facility 3      1
## 22   FRH         Facility 4    Facility 4      1
## 29   FRH         Facility 5    Facility 5      1
## 36   FRH         Facility 6    Facility 6      1
```

```
#Print final facility and model score table;
#Also print the final peer/lambda table

cat("\n")
```

```
cat("======================================================\n")
```

```
## ======================================================
```

```
cat("FINAL EFFICIENCY TABLE\n")
```

```
## FINAL EFFICIENCY TABLE
```

```
cat("======================================================\n")
```

```
## ======================================================
```

```
knitr::kable(
  efficiency_table,
  caption = "Final Efficiency Scores",
  digits = 4,
  align = "lrrrrrr"
)
```

Table 1: Final Efficiency Scores

Facility	FDH	VRS	DRS	IRS	CRS	FRH
Facility 1	1	1.0000	1.0000	1.0000	1.0000	1
Facility 2	1	1.0000	1.0000	1.0000	1.0000	1
Facility 3	1	1.0000	1.0000	1.0000	1.0000	1

Facility	FDH	VRS	DRS	IRS	CRS	FRH
Facility 4	1	1.0000	1.0000	1.0000	1.0000	1
Facility 5	1	1.0000	0.9775	1.0000	0.9775	1
Facility 6	1	0.8963	0.8675	0.8963	0.8675	1

```
cat("\n")
```

```
cat("======================================================\n")
```

```
## ======================================================
```

```
cat("FINAL PEER AND LAMBDA TABLE\n")
```

```
## FINAL PEER AND LAMBDA TABLE
```

```
cat("======================================================\n")
```

```
## ======================================================
```

```
knitr::kable(
  all_peer_lambdas,
  caption = "Final Peer and Lambda Table",
  digits = 4,
  align = c("l", "l", "l", "r")
)
```

Table 2: Final Peer and Lambda Table

	Model	Facility_Evaluated	Peer_Facility	Lambda
1	FDH	Facility 1	Facility 1	1.0000
8	FDH	Facility 2	Facility 2	1.0000
15	FDH	Facility 3	Facility 3	1.0000
22	FDH	Facility 4	Facility 4	1.0000
29	FDH	Facility 5	Facility 5	1.0000
36	FDH	Facility 6	Facility 6	1.0000
11	VRS	Facility 1	Facility 1	1.0000
6	VRS	Facility 6	Facility 1	0.4014
81	VRS	Facility 2	Facility 2	1.0000
12	VRS	Facility 6	Facility 2	0.3423
151	VRS	Facility 3	Facility 3	1.0000
221	VRS	Facility 4	Facility 4	1.0000
291	VRS	Facility 5	Facility 5	1.0000
30	VRS	Facility 6	Facility 5	0.2563
13	DRS	Facility 1	Facility 1	1.0000
5	DRS	Facility 5	Facility 1	0.2000
61	DRS	Facility 6	Facility 1	0.3429
82	DRS	Facility 2	Facility 2	1.0000
111	DRS	Facility 5	Facility 2	0.0805
121	DRS	Facility 6	Facility 2	0.3950

	Model	Facility_Evaluated	Peer_Facility	Lambda
152	DRS	Facility 3	Facility 3	1.0000
222	DRS	Facility 4	Facility 4	1.0000
23	DRS	Facility 5	Facility 4	0.5383
24	DRS	Facility 6	Facility 4	0.1311
14	IRS	Facility 1	Facility 1	1.0000
62	IRS	Facility 6	Facility 1	0.4014
83	IRS	Facility 2	Facility 2	1.0000
122	IRS	Facility 6	Facility 2	0.3423
153	IRS	Facility 3	Facility 3	1.0000
223	IRS	Facility 4	Facility 4	1.0000
292	IRS	Facility 5	Facility 5	1.0000
301	IRS	Facility 6	Facility 5	0.2563
16	CRS	Facility 1	Facility 1	1.0000
51	CRS	Facility 5	Facility 1	0.2000
63	CRS	Facility 6	Facility 1	0.3429
84	CRS	Facility 2	Facility 2	1.0000
112	CRS	Facility 5	Facility 2	0.0805
123	CRS	Facility 6	Facility 2	0.3950
154	CRS	Facility 3	Facility 3	1.0000
224	CRS	Facility 4	Facility 4	1.0000
231	CRS	Facility 5	Facility 4	0.5383
241	CRS	Facility 6	Facility 4	0.1311
17	FRH	Facility 1	Facility 1	1.0000
85	FRH	Facility 2	Facility 2	1.0000
155	FRH	Facility 3	Facility 3	1.0000
225	FRH	Facility 4	Facility 4	1.0000
293	FRH	Facility 5	Facility 5	1.0000
361	FRH	Facility 6	Facility 6	1.0000

The DEA results indicate that Facilities 1–4 are efficient under all six assumptions. Facility 5 is inefficient under DRS and CRS, with an efficiency score of 0.9775, which means they could reduce input by 2.3% and produce the same output if operating under assumptions of decreasing returns to scale or constant returns to scale . Facility 6 is the least efficient, scoring 0.8963 under VRS and IRS and 0.8675 under DRS and CRS. That means facility 6 could reduce inputs by 10.4% or 13.3% depending on which assumptions are made about returns to scale. FDH and FRH classify all facilities as efficient. Overall, CRS and DRS identify greater inefficiency than VRS and IRS, while FDH and FRH provide no differentiation among facilities in this dataset.